

Series 2: Applied Machine Learning and Predictive Modelling 1

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Exercise

In this exercise we are going to use the “Ozone” data set from the `gss` (type `?gss::ozone` for more information about the data).

```
## if not yet there, install the package first:
# install.packages("gss")
data(ozone, package = "gss")
##
head(ozone)
```

	upo3	vdht	wdsp	hmdt	sbtp	ibht	dgpg	ibtp	vsty	day
1	3	5710	4	28	40	2693	-25	87	250	3
2	5	5700	3	37	45	590	-24	128	100	4
3	5	5760	3	51	54	1450	25	139	60	5
4	6	5720	4	69	35	1568	15	121	60	6
5	4	5790	6	19	45	2631	-33	123	100	7
6	4	5790	3	25	55	554	-28	182	250	8

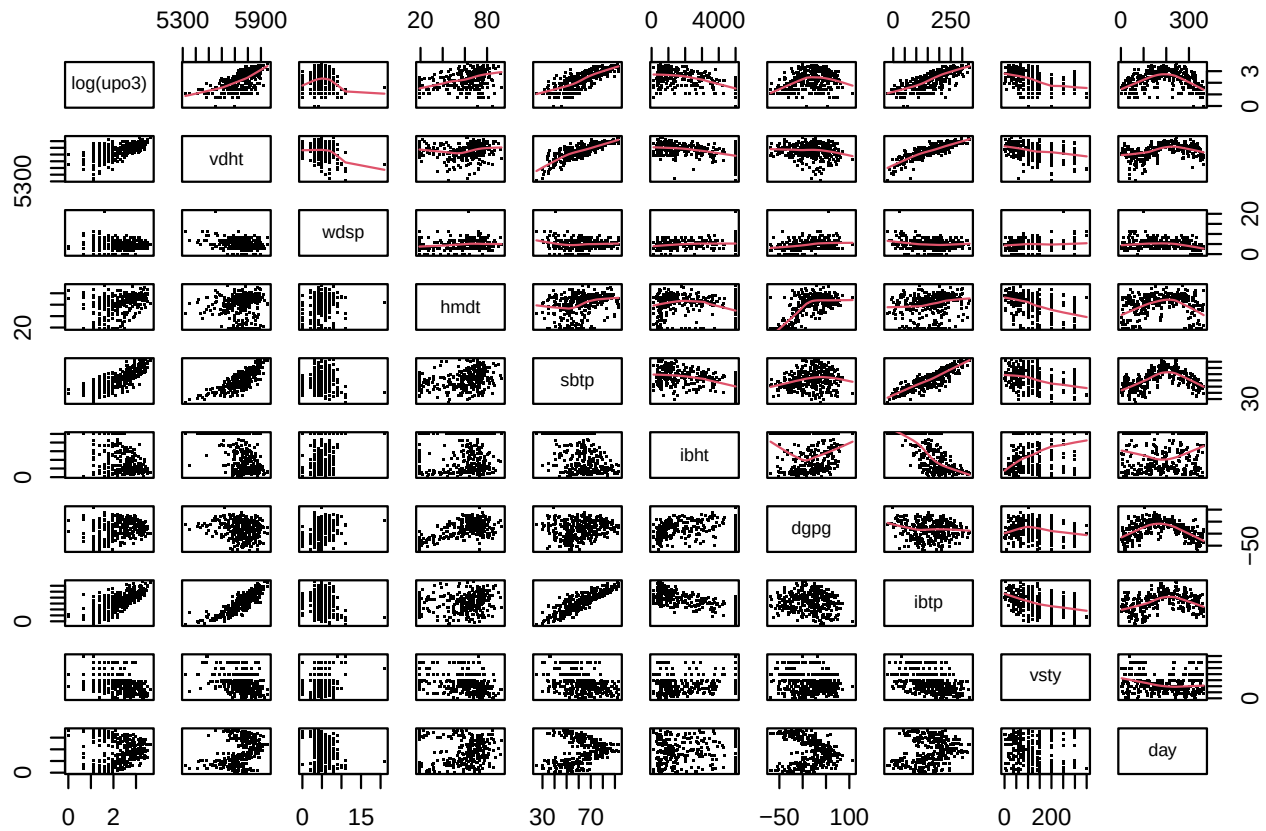
```
# ?gss::ozone
```

Our aim here is to fit a GAM model to the `ozone` data set.

As usual we start by visualising the response variable against each predictor. As the response variable can only take positive values (i.e. is an “amount”) we log-transform it. This is often the standard procedure with concentrations.

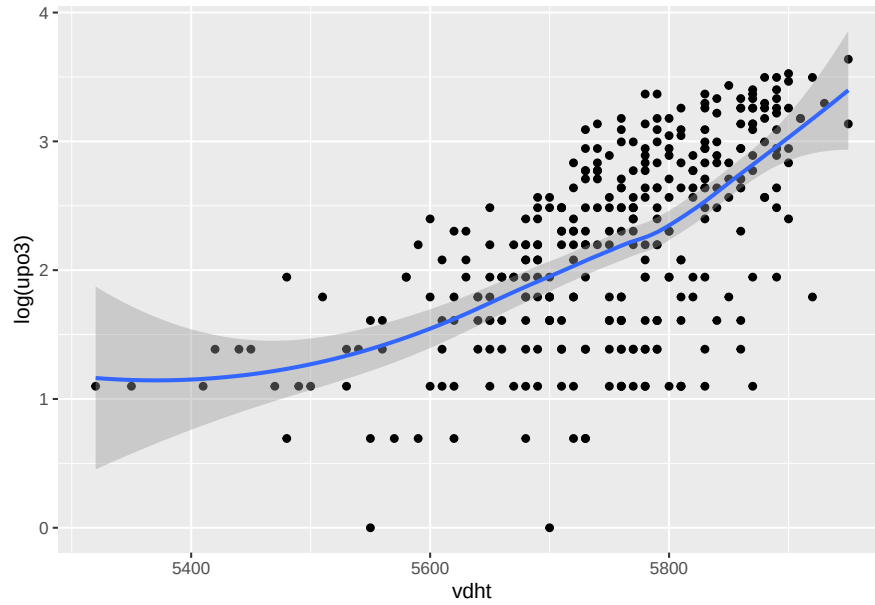
To start “quick and dirty” we can create a graph with `pairs()`.

```
pairs(log(upo3) ~ . ,
      data = ozone,
      pch = ".",
      upper.panel = panel.smooth)
```



As there are many predictors, the single graphs are too small. Therefore, we prefer to create each graph separately. Let's start with `vdht`.

```
library(ggplot2)
ggplot(data = ozone,
       mapping = aes(y = log(upo3),
                     x = vdht)) +
  geom_point() +
  geom_smooth()
```



This relationship appears to be non-linear. It is possible that a simple quadratic term is enough to model it. Nevertheless, we will fit a GAM model and then look at the estimated smooth function and at its “edf” (i.e. estimated degrees of freedom) that quantify the complexity of a smooth term.

Question

Go through all the other predictors, produce the corresponding graph and comment on the relationship. You may want to say whether the relationship can be assumed to be linear, quadratic or more complex. If you prefer, you may look at a couple of predictors only (as there are many).

Question

Fit a Generalised Additive Model with the `gam()` function from `{mgcv}` package. Remember to fit the model to the log-transformed response variable.

Question

Look at the summary of the model you just fitted and answer the following questions: i) How many of the smooth terms appear to have a significant effect if we were to use a strict 5% threshold? ii) Which smooth terms are estimated to have linear effect? iii) Which one is the term that is estimated to be the most complex? iv) Are there any "parametric" terms in this model?

Question

Use the `plot()` function to visualise the estimated smooth terms.

Question

Ask generative AI to

- *provide an example or an explanation when it is important to include polynomial effects in a linear model*
- *explain the concept of collinearity*
- *explain when it is appropriate to log-transform a variable before putting it in a model*

and compare it with the definitions you find in the lecture materials. Do you think they are different in any way? Which one is easier for you to understand?